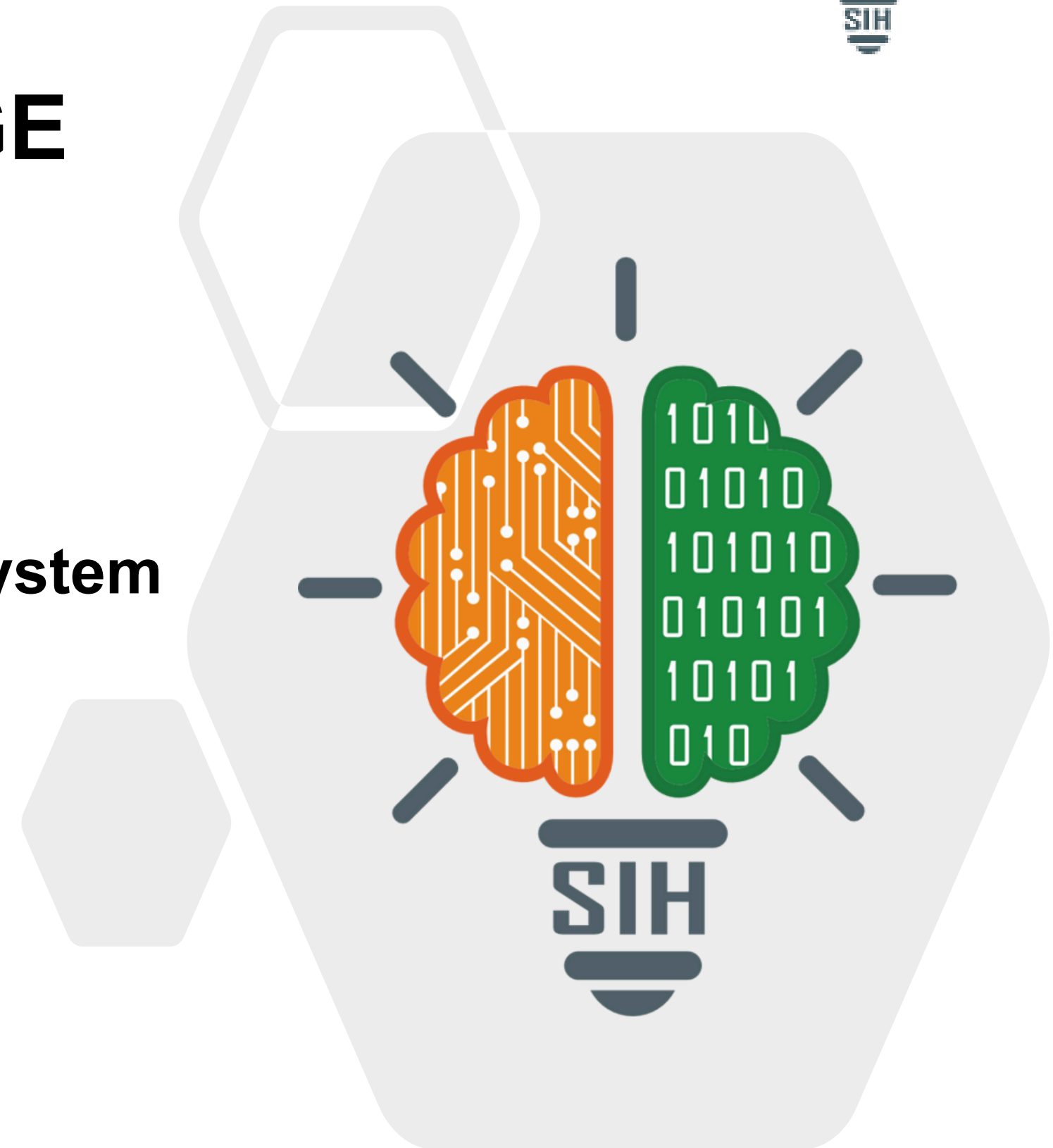


TITLE PAGE

- Problem Statement ID – SIH25102
- Problem Statement Title- AI-based Drop-out Prediction and Counseling System
- Theme- Smart Automation
- PS Category- Software
- Team ID-
- Team Name- Impact Crew



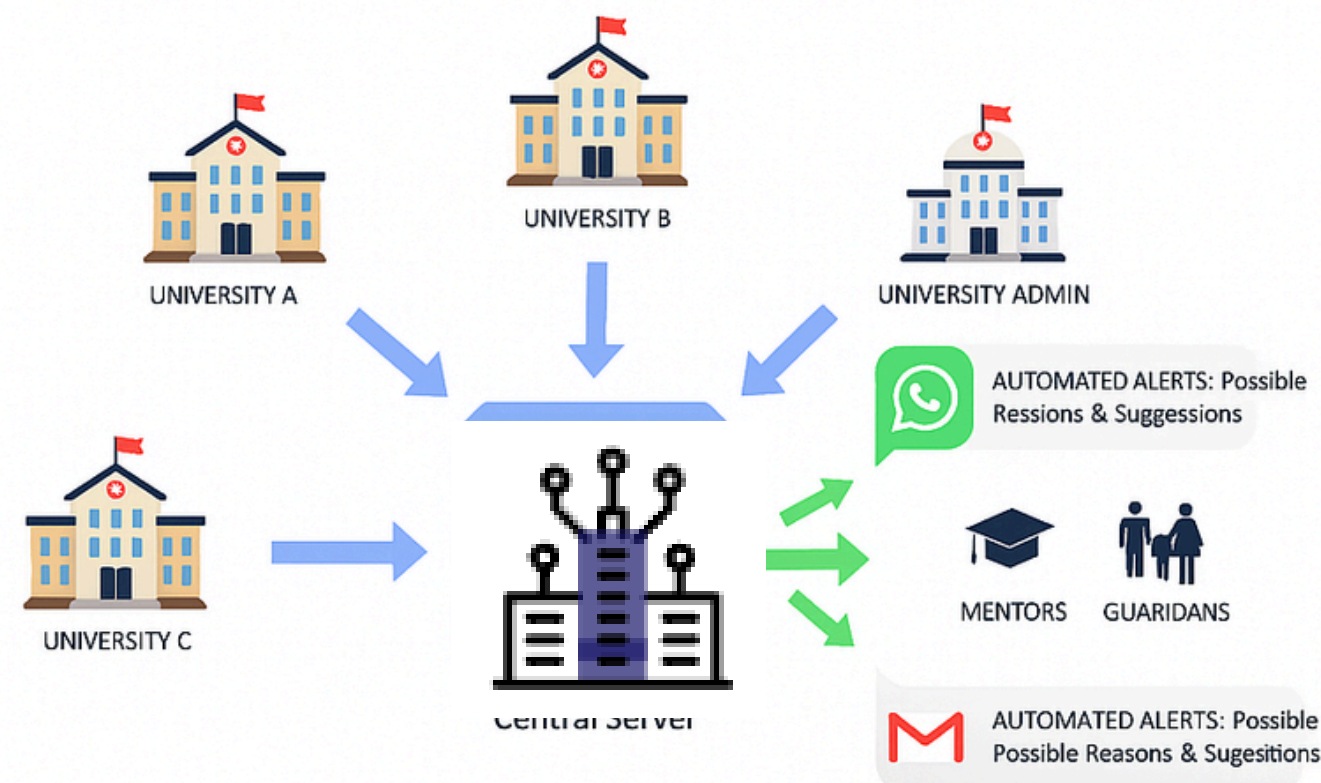
Actionable Insights & Counselling Support

Colour-coded dashboards, student timelines, and automated **alerts** guide mentors and guardians to intervene with clear, data-driven **reasoning**.



AI-Powered Prioritisation

Lightweight, **explainable ML** ranks flagged students by **urgency**, helping mentors **act faster** on those most likely to drop out.



Early Risk Detection

Smart, rule-based analytics continuously monitor attendance %, test score trends, and fee status to **flag at-risk students** before final exams expose failures.

Centralized Web Platform



A unified website that **merges** attendance, academics, and fee records, solving the issue of **scattered data**

Frontend

React, Tailwind CSS, Radix UI, Lucide React, Axios, GSAP, Framer Motion, Recharts

Backend

Node.js, MongoDB, JWT, bcryptjs, Papaparse

AI & ML

Pandas, NumPy, Scikit-learn, XGBoost, TensorFlow/PyTorch

API Services & Notifications

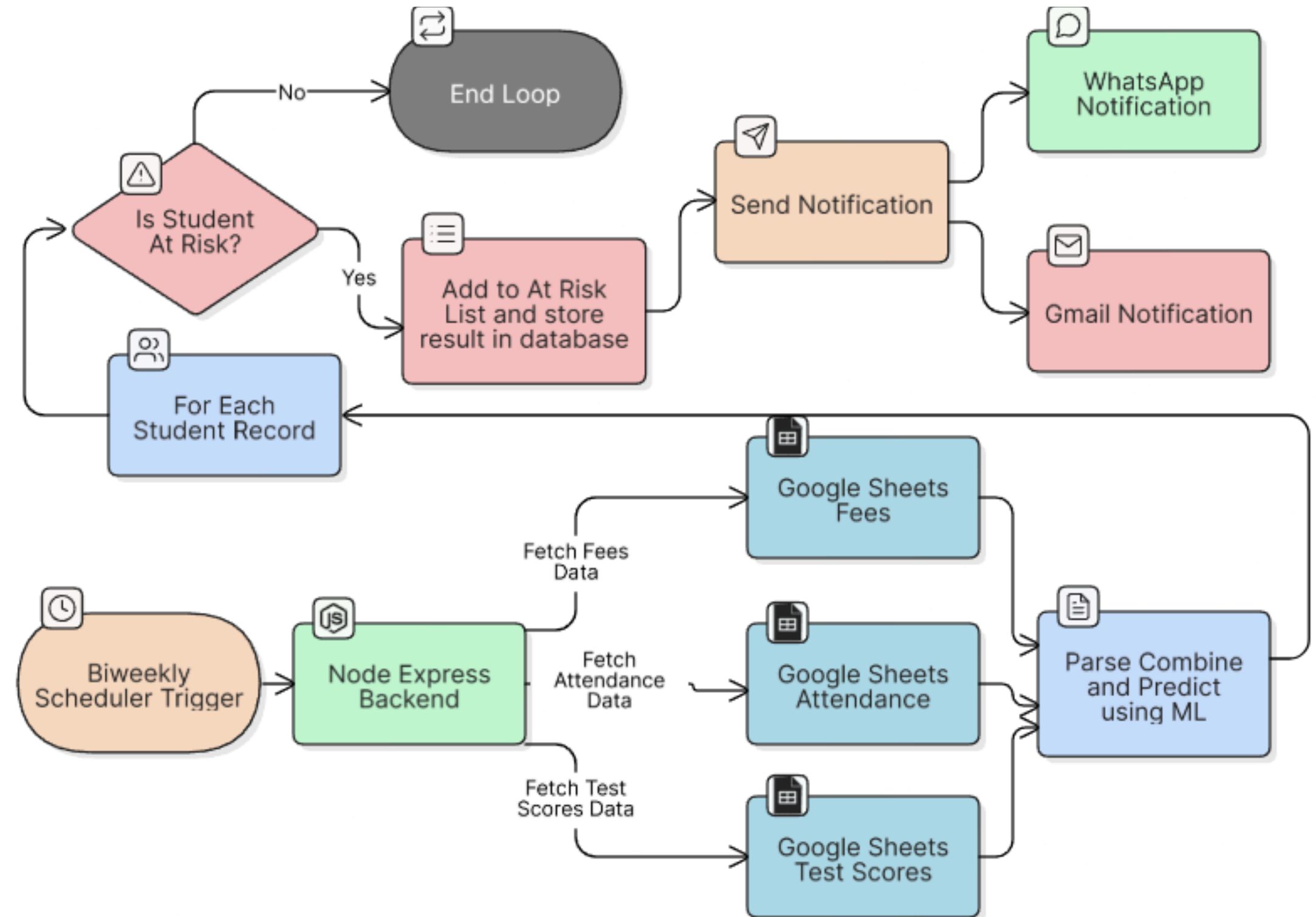
SMTP, Twilio

Deployment

Vercel, Render, GitHub, Cron Jobs

Developer Tools

GitHub, Postman, Jupyter Notebook, VS Code



Implement mitigation strategies to overcome challenges

Optimize API usage, apply automated data-cleaning processes

Address API limitations and data inconsistencies

API limits may restrict data fetch frequency & data inconsistencies necessitate robust cleaning routines to ensure data quality.

Ensure reliable notification delivery

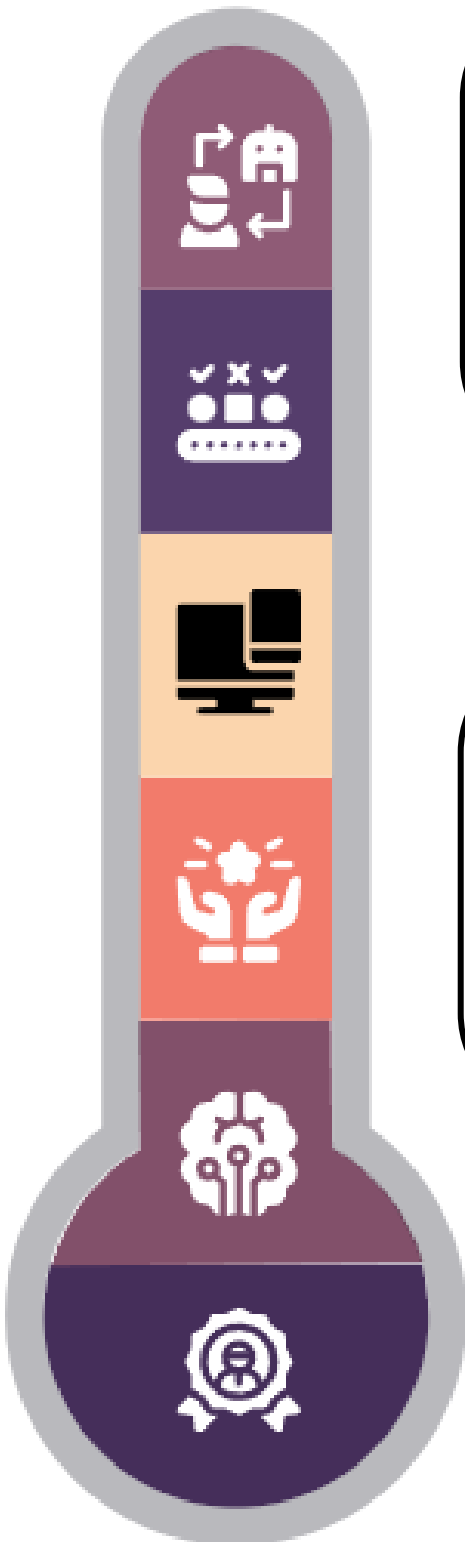
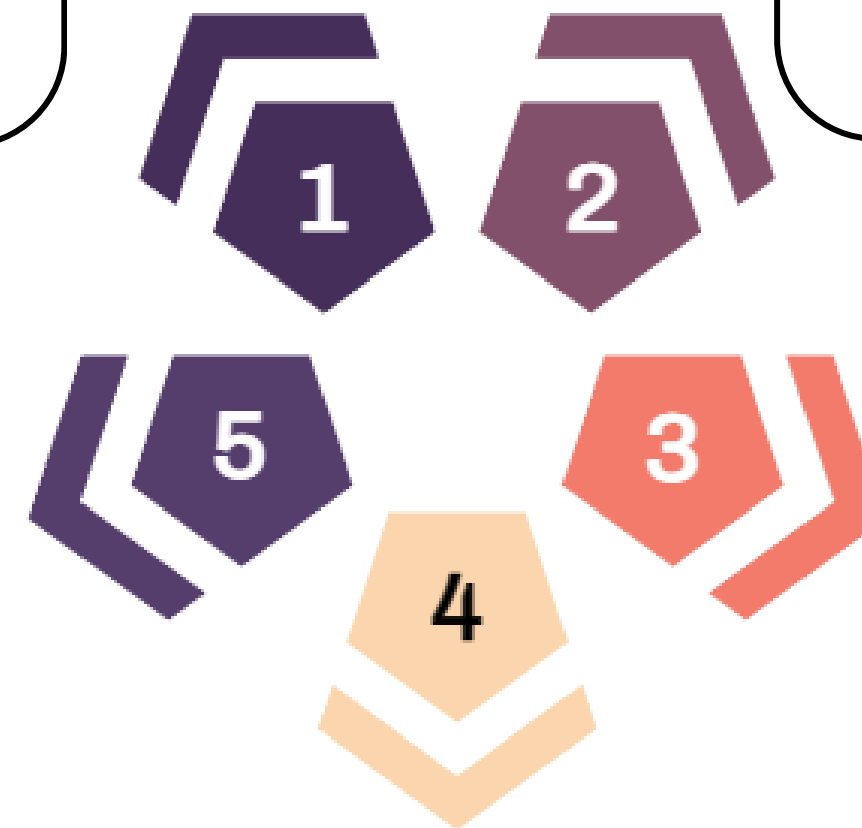
Challenges include maintaining dependable delivery of notifications via WhatsApp and Gmail, critical for timely interventions.

Leverage existing infrastructure for feasibility

Utilizes Google Sheets, free APIs, and open-source tools to minimize upfront costs and enable rapid deployment.

Utilizing Existing Infrastructure for Feasibility

By integrating Google Sheets, free APIs, and open-source tools, we can minimize initial costs and facilitate swift deployment.



Innovation and Uniqueness of Our Solution



Offer a budget-friendly alternative to costly analytics platforms

The solution is designed to be cost-effective compared to expensive analytics tools, making it accessible for educational institutions with limited budgets.



Leverage widely available tools like Google Sheets and Node.js

Built on common platforms such as Google Sheets for data input and Node.js for backend processing, ensuring ease of adoption and maintenance.



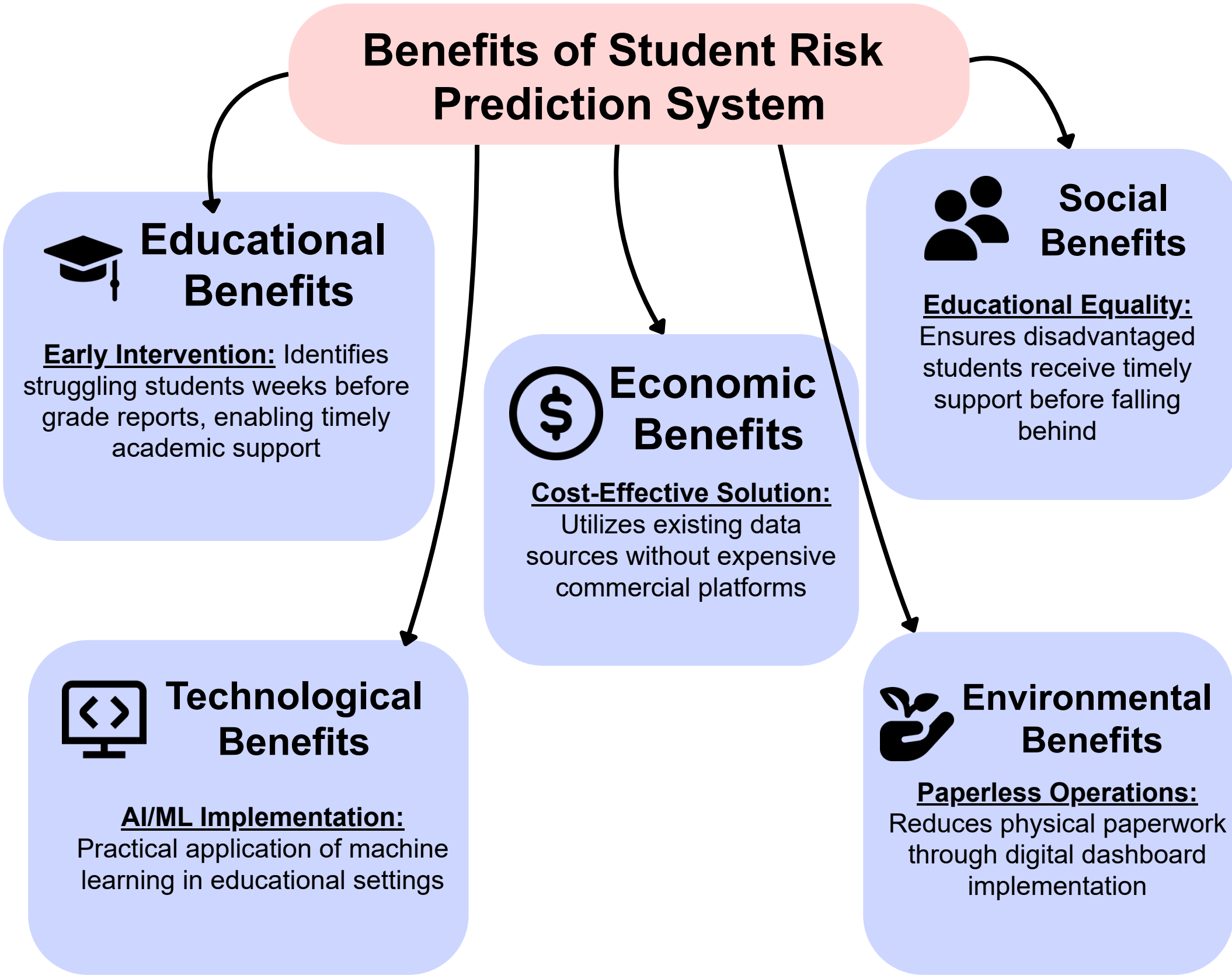
Focus on early alerts and empower educators without replacing judgment

Emphasizes providing timely notifications and insights to mentors, supporting human decision-making rather than automating it fully.



Ensure transparency, easy configuration, and high scalability

The system is designed to be clear in its operations, simple to configure by users, and scalable to handle growing data and user demands.



Research Sources

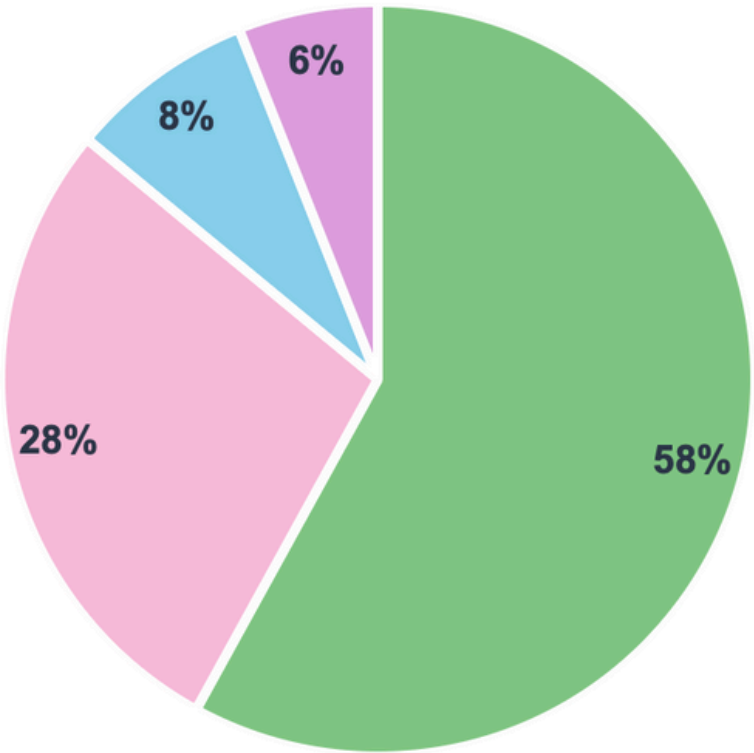
<https://ijabs.niilmuniversity.ac.in/exploring-the-factors-affecting-dropout-rates-in-higher-education-institutions-in-india/>
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9980764/>
<https://educationforallinindia.com/dropout-rates-in-schools-in-india/>

ML Dataset

<https://www.kaggle.com/datasets/thedevastator/higher-education-predictors-of-student-retention>

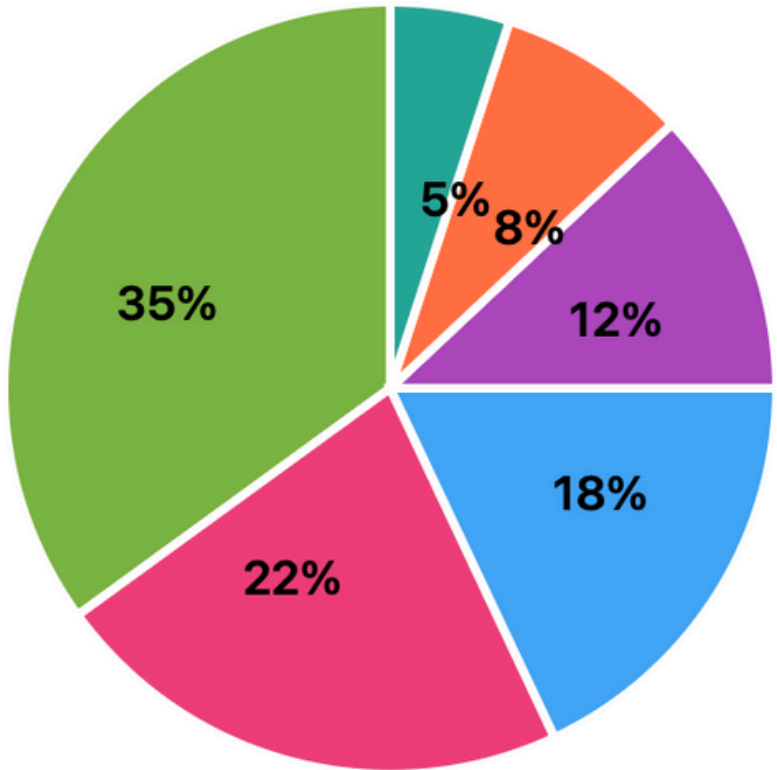
ML Training Model

<https://arxiv.org/abs/1603.02754>



● Successfully Graduated (58%) ● Dropped Out (28%)
● Academic Break/Gap Year (8%)
● Still Enrolled Beyond Duration (6%)

Data based on All India Survey on Higher Education (AISHE) 2021-22 and Ministry of Education statistics for undergraduate programs in Indian universities



● Financial Constraints (35%)
● Family Pressure & Marriage (22%)
● Poor Academic Performance (18%)
● Employment Opportunities (12%)
● Language & Cultural Barriers (8%)
● Health Issues (5%)

Data based on All India Survey on Higher Education (AISHE) 2021-22 and Ministry of Education statistics for undergraduate programs in Indian universities